

# KYURI PARK

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## EDUCATION

|                                                                                                                                                                                                            |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| University of Amsterdam - Computational Science Lab, Informatics Institute                                                                                                                                 | Aug 2023-Present |
| <ul style="list-style-type: none"><li>PhD Candidate; expected completion August 2027.</li><li>Supervisors: Vitor V. Vasconcelos and Mike Lees.</li></ul>                                                   |                  |
| Visiting Scholar, Stanford University - Stanford Doerr School of Sustainability                                                                                                                            | Spring 2026      |
| <ul style="list-style-type: none"><li>Hosted by Sara Constantino; research on climate-attitude networks, belief systems, social norms, and exploratory causal discovery.</li></ul>                         |                  |
| Complex Systems Summer School, Santa Fe Institute                                                                                                                                                          | Summer 2026      |
| <ul style="list-style-type: none"><li>Selected participant; interdisciplinary training, collaborative modeling projects, and community outreach.</li></ul>                                                 |                  |
| MSc, Methodology and Statistics, Utrecht University                                                                                                                                                        | Jun 2023         |
| <ul style="list-style-type: none"><li>Cum laude, GPA 9.04/10.00.</li><li>Supervisors: Oisín Ryan and Lourens J. Waldorp.</li></ul>                                                                         |                  |
| BSc, Psychology (Psychological Methods), University of Amsterdam                                                                                                                                           | Jun 2021         |
| <ul style="list-style-type: none"><li>Cum laude, GPA 9.05/10.00; Honors Programme.</li><li>Supervisors: Claudia van Borkulo and Denny Borsboom.</li></ul>                                                  |                  |
| BSc, Business Management, University of Houston                                                                                                                                                            | May 2017         |
| <ul style="list-style-type: none"><li>Summa cum laude, GPA 3.94/4.00; specialization in Finance Management and Accounting.</li><li>Exchange semester at Mount Royal University, Calgary, Canada.</li></ul> |                  |

## RESEARCH INTERESTS

Complex systems | Causal discovery | Dynamical systems | Network analysis | Computational modeling | Multiscale dynamics | Mental health | Climate beliefs, behavior, and adaptation

## PEER-REVIEWED PUBLICATIONS

- Park, K., Waldorp, L. J., & Vasconcelos, V. V. (2026). A mechanistic model of symptom dynamics: Implications for statistical network analyses. *Multivariate Behavioral Research*. Advance online publication. doi:10.1080/00273171.2026.2710730.
- Park, K., Vasconcelos, V. V., & Lees, M. (2026). A tutorial on causal network simulation and exploration using the *causalnet* R package. *Behavior Research Methods*, 58, 204. doi:10.3758/s13428-026-03030-z.
- Park, K., Li, X., Waldorp, L. J., Lees, M., & Vasconcelos, V. V. (2026). The role of feedback loops in dynamical symptom networks. *Scientific Reports*, 16, 11273. doi:10.1038/s41598-026-38747-6.
- Park, K., Elsenburg, L. K., Nicolaou, M., Stronks, K., & Vasconcelos, V. V. (2026). Connecting precariousness and depression: From causal discovery to intervention simulation. *SSM - Mental Health*, 9, 100637. doi:10.1016/j.ssmmh.2026.100637.
- Elsenburg, L. K., Stronks, K., Galenkamp, H., Lakerveld, J., Lok, A., Park, K., Vasconcelos, V. V., & Nicolaou, M. (2026). Precariousness and depressed mood: A network analysis in the multi-ethnic HELIUS study. *Social Psychiatry and Psychiatric Epidemiology*. doi:10.1007/s00127-026-03072-w.
- Park, K., Waldorp, L. J., & Ryan, O. (2024). Discovering cyclic causal models in psychological research. *advances.in/psychology*, 2(1), e72425. doi:10.56296/aip00012.

### Preprint

- Lee, S. Y., Hwang, H., Kim, T., Kim, Y., Park, K., Yoo, J., Borsboom, D., & Shin, K. (2025). Emergence of psychopathological computations in large language models. arXiv. doi:10.48550/arXiv.2504.08016. Under review at *Nature Communications*
- Park, K., Borsboom, D., Lees, M., Elsenburg, L., Lunansky, G., Stronks, K., ... Vasconcelos, V. (2026). Rethinking Group Differences in Psychopathology Networks: A Slow-Fast Perspective on Context and Symptom Activation. [osf.io/preprints/psyarxiv/u8wf5\\_v1](https://osf.io/preprints/psyarxiv/u8wf5_v1). Under review at *Nature Mental Health*

### Selected research in progress

- Time-scale separation in dynamical models of psychological processes.
- Computational modeling of executive-function engagement across development.
- Comparative climate-attitude networks and exploratory causal discovery across Japan, Europe, and the United States.
- Institutionalized category convergence in heterogeneous rate-distortion agent-based models.
- Post-disaster population displacement, migration, and urban adaptation.

## GRANTS, FELLOWSHIPS, AND AWARDS

- Van der Gaag Grant, Royal Netherlands Academy of Arts and Sciences (KNAW) - PI, 2026-2027.
- Bridge Grant, Complex Systems Society, 2026.
- Energy Transition through the Lens of the SDGs (ENLENS) Research Grant, 2025-2027.
- Travel Award, International Meeting of the Psychometric Society, 2025 and 2026.
- Amsterdam University Fund - PhD Grant, 2025.
- Utrecht Excellence Scholarship, Utrecht University, 2021-2023.
- Scholarship Excellence Award, University of Houston, 2017
- UH Academic Excellence Scholarship, University of Houston, 2014-2017.

## SELECTED PRESENTATIONS

- Park, K. (Jul 2026). A slow-fast perspective on between-group differences in symptom networks. International Meeting of the Psychometric Society, Seoul, South Korea.
- Park, K. (Jul 2025). Computational modeling of symptom dynamics and their implications for statistical networks. International Conference on Computational Science, Singapore.
- Park, K. (Jul 2025). Unraveling symptom dynamics: A mechanistic approach to feedback loops and causal discovery. International Meeting of the Psychometric Society, Minneapolis, MN.

INVITED TALKS AND SEMINARS

- Park, K. (May 2026). Executive function engagement as a slow-fast stochastic dynamical system. Center for Mind and Brain, University of California, Davis, Davis, CA, USA.
- Park, K. (Apr 2026). Complex systems approaches to mental health and climate-change attitudes. Biohealth Convergence and Open Sharing System (COSS), South Korea.
- Park, K. (Feb 2026). Slow-fast coupling: Curie-Weiss symptoms under context dynamics. 3rd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.
- Park, K. (Jan 2026). Structural risk shifts symptom thresholds without altered symptom coupling: A slow-fast perspective. Theory Methods Lab Seminar, Amsterdam.
- Park, K. (Mar 2025). Computational approaches to mental health at the symptom level. 2nd Joint Kobe University-UvA Workshop on Computational Social Science and Intelligent Systems, Amsterdam.

RESEARCH EXPERIENCE

- University of Amsterdam - Computational Science Lab, Informatics Institute** Aug 2023-Present
- Develop causal and dynamical models of complex human systems, with applications in mental health, climate attitudes, belief dynamics, and urban adaptation.
  - Combine causal discovery, network analysis, agent-based modeling, and simulation to study feedback processes, structural uncertainty, and intervention effects.
  - Lead developer of `causalnet`, an open-source R package for generating alternative directed structures from a network skeleton, including networks with cycles, summarizing their properties, and simulating nonlinear or user-defined dynamics.

- Utrecht University** Jun 2021-Aug 2023
- Enhanced the R package `mHMMbayes` (Multilevel Hidden Markov Models Using Bayesian Estimation) and developed tutorial materials.
  - Contributed to work on the `mice` R package and developed a synthetic-data approach for evaluating imputation-model validity with Gerko Vink.
  - Developed procedures for producing reproducible synthetic-data extracts from Statistics Netherlands data and co-developed the `synthpop.extract` R package with Erik-Jan van Kesteren.
  - Translated a Bayesian piecewise-linear regression implementation for dynamic network construction from MATLAB to R.
  - Applied and compared cyclic causal-discovery algorithms (CCD, FCI, CCI) for psychological data.
  - Coded methodological characteristics of psychological network studies in NVivo with Noémi Schuurman.

- University of Amsterdam - Psychological Methods** Nov 2019-Jan 2021
- Conducted a meta-analysis of predictors of science skepticism, developed `JASP` tutorials, and modeled networks containing variables operating at different timescales using Ising and mixed graphical models.

TEACHING AND STUDENT MENTORING

- University of Amsterdam**
- Teaching Assistant, *Complexity: Can It Be Simplified?* (BSc Honors/MSc), Vítor V. Vasconcelos, 2023–2025.
  - Teaching Assistant, *Causality* (MSc Artificial Intelligence), Sara Magliacane, 2024–2025.
  - Teaching Assistant, *Modeling System Dynamics* (MSc Information Studies), Debraj Roy, 2023–2024.
  - Teaching Assistant, *AI and Psychology* (BSc Psychology), Han van der Maas, 2021.

- Utrecht University**
- Developed practical materials for Python programming, Bayesian networks, relational-event models, missing data, imputation, and text mining courses; provided R programming support to sociology graduate students.

- University of Pennsylvania**
- Teaching Assistant, *Social Economics: An Economic Perspective on Human Behavior*, Femida Handy, Summer 2016.

- MSc Student Supervision**
- **Henry Zwart**, MSc in Computational Science (2026). Thesis: *Interventions on Asymmetric Belief Networks*.
  - **Paraskevas Leivadarios**, MSc in Data Science (2025). Thesis: *Elections and climate attitudes: how do people's views on climate change and related policies change during an election?*
  - **Gabriela Torres**, MSc in Data Science (2025). Thesis: *The Impact of COVID-19 Interventions and Extreme Weather Harm on Support for Climate Policy*.
  - **Xinhai Li**, MSc in Computational Science (2024). Thesis: *Feedback Loops in the Dynamics of Major Depressive Disorder*.

ADDITIONAL EXPERIENCE

- SERVICE AND LEADERSHIP**
- POLDER, University of Amsterdam** Sep 2026-Present
- **Research Coordinator**; organize interdisciplinary events and coordinate the POLDER research community across complexity science, computational modeling, and policy.
- Utrecht University - Human Data Science Group** 2021-2023
- Organized research and data-science reading groups, helped coordinate Data Donation Day, and maintained the group website.
- University of Amsterdam** 2018-2021
- International Student Ambassador and member of the Psychological Methods Student Faction; represented students in programme meetings and organized academic, orientation, and outreach events.

- PROFESSIONAL EXPERIENCE**
- Emirates Airline, Dubai, UAE** May 2017-Jul 2018
- Cabin crew
- Amerril Energy LLC, Houston, TX, USA** Jun 2016-Apr 2017
- Accounting assistant

METHODS, PROGRAMMING, AND LANGUAGES

**Methods:** Causal discovery, network analysis, dynamical-systems modeling, stochastic differential equations, agent-based modeling, simulation studies, psychometric modeling, and Bayesian and multilevel modeling.  
**Programming:** R, Python, MATLAB; JavaScript (learning), C++ (learning)  
**Languages:** Korean (native), English (fluent), Dutch (B1).